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RESEARCH ARTICLE

TrBot: A Turkish Deep Learning Chatbot Utilizing Seq2Seq Model

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ABSTRACT In Natural Language Processing (NLP) and Artificial Intelligence (AI), chatbots, which are software programs designed to facilitate human-computer interaction through natural language, are becoming increasingly important. However, creating an effective chatbot remains a complex task, as it must accurately interpret user input and generate appropriate responses. This study presents TrBot, a general-purpose Turkish chatbot that utilizes deep learning techniques, specifically a seq2seq model with Long Short-Term Memory (LSTM) layers. This architecture allows TrBot to manage sequential dependencies and effectively generate coherent responses, offering advantages in handling the complex morphological structure of Turkish. In contrast to earlier Turkish chatbots that were application-specific, TrBot is designed for broad conversational use across various topics. In this study, we also proposed and created two comprehensive datasets: a QA dataset with 40,702 entries and a conversation dataset with 304,446 entries, both specifically designed to enhance TrBot's performance. Trained on these datasets, TrBot achieved an accuracy of 80% on the QA dataset and 70% on the dialog dataset, with BLEU scores of 0.90 and 0.77 respectively, indicating substantial enhancements in response quality. In comparison, a Transformer-based model exhibited reduced training times but achieved lower accuracies of 60% on the QA dataset and 50% on the dialog dataset, with BLEU scores of 0.76 and 0.61 respectively, with the limited size of the datasets and available computational resources. The development of TrBot has significant implications, offering potential benefits in areas such as customer support, language learning, and other fields that require robust Turkish conversational capabilities. This study demonstrates that with adequate data and appropriate modeling techniques, it is possible to create effective conversational agents for complex languages like Turkish, paving the way for further advancements in this domain.

INDEX TERMS Chatbots, deep learning, LSTM, natural language processing, Seq2seq model, Turkish language.

I. INTRODUCTION

Chatbots, also known as conversational agents, have become integral tools for enhancing human-computer interactions across a wide range of applications [1]. These systems are designed to comprehend natural language inputs and generate appropriate responses, making them indispensable tools across multiple industries due to their ability to provide instant, round-the-clock customer support, streamline business processes, and enhance user engagement [2], [3]. They are widely used in customer service to handle inquiries, troubleshoot problems, and facilitate transactions, thereby improving customer satisfaction and operational efficiency [4]. In healthcare, chatbots assist in patient monitoring, appointment scheduling, and providing medical information, making healthcare more accessible and efficient [5], [6]. In education, they offer personalized tutoring, administrative support, and student engagement, promoting interactive and adaptive learning experiences [7]. Additionally, chatbots play a crucial role in e-commerce by guiding users through

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their shopping journey, recommending products, and processing orders, which boosts sales and customer retention [8].

The core of chatbot effectiveness lies in conversation modeling, a specialized field that intersects with both Natural Language Processing (NLP) and Artificial Intelligence (AI) [9]. By leveraging advanced techniques like deep learning, chatbots aim to emulate human-like conversational behavior, thereby significantly improving user engagement and satisfaction.

Despite the rapid advancements in AI, developing a highly effective chatbot remains a formidable challenge. A key objective in chatbot research is to enable these systems to accurately comprehend user intent and respond appropriately in real-time. Achieving this objective requires sophisticated algorithms capable of processing complex language structures and understanding contextual nuances.

In recent years, sequence-to-sequence (Seq2Seq) models have become a leading approach for conversation modeling [10]. These models, typically based on recurrent neural networks (RNNs), excel at capturing sequential dependencies and generating coherent responses. Trained on large-scale datasets containing question-answer pairs or conversational exchanges, seq2seq models can effectively simulate human-like conversations by mapping input sequences to output sequences. Transformer models, which rely on attention mechanisms rather than recurrence, have further revolutionized the field [11]. Transformers enable the parallel processing of input data, leading to more efficient training and improved handling of long-range dependencies. Their ability to focus on different parts of the input sequence simultaneously allows for more contextually relevant responses.

The exploration of chatbots within the Turkish language context highlights a significant research gap. Despite the advancements in chatbot technologies, there is a notable scarcity of studies focusing on Turkish conversational agents, particularly those that tackle the language's inherent complexity and the limited availability of large-scale, comprehensive datasets. The Turkish language, with its complex morphological structures, presents unique challenges for NLP models. Existing Turkish chatbots are often application-specific and lack the versatility needed for general conversational contexts. This gap emphasizes the necessity for a robust investigation aimed at developing a versatile, deep learning-based Turkish chatbot. Such a chatbot should adeptly handle the unique morphological features of Turkish and provide meaningful interactions across a broad range of applications.

The adoption of deep learning has facilitated the creation of language-specific chatbots tailored to diverse linguistic contexts. This paper introduces TrBot, a deep learning-based Turkish chatbot developed for general context use. TrBot exemplifies the trend of leveraging seq2seq model architectures to engage in text-based conversations in the Turkish language. By training on datasets comprising question-answer pairs and conversational interactions, TrBot addresses the linguistic complexities of Turkish morphology and demonstrates the potential of deep learning in creating meaningful user interactions. TrBot represents a significant advancement in Turkish chatbots, particularly given the complexity of the Turkish language, which is characterized by intricate morphological structures. Turkish is among the top ten most spoken languages in terms the numbers of speakers and area, and is used by around 200 million people [12]. The proliferation of Turkish-language websites and blogs further underscores the language's versatility and significance. As such, Turkish chatbots like TrBot have the potential to handle a diverse range of tasks, from customer support to language instruction, effectively facilitating meaningful dialogues across a broad spectrum of topics. Accordingly, the main contributions of this study can be summarized as follows:

- Introduction of TrBot, the general-purpose Turkish chatbot.
- Development and utilization of two comprehensive datasets created explicitly for this study: a QA dataset with 40,702 entries and a conversation dataset with 304,446 entries.
- Implementing of a Seq2Seq model with Long Short-Term Memory (LSTM) layers, effectively capturing sequential dependencies and generating coherent responses.
- Development and evaluation of an additional version of TrBot using transformer-based architecture for comparison. Our findings indicate that while the transformer-based model shows promise, the seq2seq model outperforms it in handling the Turkish language's complex morphological structure.

The remainder of this paper is structured as follows: Section II provides a comprehensive review of existing literature on chatbot development methodologies, with a particular emphasis on deep learning approaches and seq2seq models. Section III outlines the methodology used in the design and training of TrBot, covering dataset creation, model architecture, and training procedures. Section IV presents the experimental setup and results obtained from evaluating TrBot's performance. Section V then discusses the implications of these findings, explores potential applications of TrBot, and suggests avenues for future research. Finally, Section VI concludes the paper by summarizing the key insights and contributions to the field of chatbot development.

II. LITERATURE REVIEW

The rapid evolution of chatbot technology has been driven by the increasing demand for natural language interaction between humans and machines. As articulated in Turing's seminal work, the quest to create intelligent machines capable of engaging in human-like conversation has been a longstanding endeavor [13]. The emergence of chatbots represents a significant milestone at the intersection of NLP and AI.

Chatbot systems based on rule-based approaches demonstrated the feasibility of simulating conversation [14], [15], [16]. However, the limitations of these rule-based systems in handling complex linguistic patterns and understanding user intent necessitated the exploration of more sophisticated techniques. Caldarini et al. [17] differentiate between rule-based chatbots, such as Eliza, and AI-based chatbots, like ChatGPT. Rule-based chatbots, which rely on pattern matching, are simpler to implement but struggle with complex queries. AI-based chatbots, on the other hand, utilize deep learning and NLP techniques, which allow them to learn from large datasets of human interactions and provide more sophisticated responses.

The introduction of deep learning models, particularly Seq2Seq architectures, marked a paradigm shift in chatbot development [10]. These models leverage neural networks to learn from large-scale datasets, enabling them to generate contextually relevant responses to user queries.

The literature identifies several types of chatbots based on criteria such as interaction mode, knowledge domain, and goals. Studies [17], [18], [19], [20], [21], [22], and [23] categorize chatbots into various types. Hussain et al. [18] classify chatbots into task-oriented and non-task-oriented categories. Task-oriented chatbots assist users in completing specific tasks, such as booking a flight or making a reservation, with examples including Cortana, Alexa, and Siri. Conversely, non-task-oriented chatbots are designed to engage in broader conversations without specific goals.

Braun and Matthes [19] distinguish between open-domain chatbots, which can converse on a wide range of topics, and closed-domain chatbots, which focus on a specific knowledge area. Chatbots are further divided into proactive and reactive types [20], [24]. Proactive chatbots offer assistance without requiring explicit user requests, such as sending notifications or alerts, while reactive chatbots respond to user-initiated queries, like booking services. Proactive and reactive features are often integrated, allowing chatbots to provide a comprehensive range of interactions. For example, after a reactive reservation, a chatbot might proactively suggest additional activities, or it could alert users about weather conditions during a trip.

Recent advancements in chatbot technology have explored the integration of personality and identity models to enhance user engagement and immersion [25]. These approaches allow chatbots to simulate the voices and personalities of fictional characters or real-life personas, creating more compelling and authentic conversational experiences. Additionally, the adoption of reinforcement learning techniques has facilitated dynamic adaptation and learning from user interactions, further improving chatbot performance and responsiveness [26].

In the educational domain, chatbots have emerged as valuable tools for personalized learning and assessment. Sequence-based neural models, as proposed by [27], offer insights into student learning outcomes and provide tailored feedback to enhance the learning experience. By leveraging machine learning techniques, educational chatbots can adapt to individual learning needs and foster student engagement and achievement.

Transformer models introduced by Vaswani et al. [11] have redefined state-of-the-art in NLP by introducing attention mechanism and parallel processing capabilities make them efficient for handling long-range dependencies in language tasks. Bidirectional Encoder Representations from Transformers (BERT) is a novel NLP paradigm that has made important contributions to language interpretation and text analysis [28]. BERT's architecture, developed by Google AI researchers in 2018, is based on the transformer model and uses an attention mechanism. This attention mechanism enables BERT to focus on different portions of the input sentence while encoding each word or token leading to better language understanding. In 2022, ChatGPT marked a significant advancement in chatbot technology, built on the GPT language model [29]. Developed by OpenAI, it is a highly sophisticated model designed to assist with tasks such as generating human-like responses, answering questions, and providing information on various topics. Shortly thereafter, Gemini, formerly known as Bard and developed by Google, emerged. Gemini is gaining popularity for its speed and capability to respond in a human-like manner by accessing current information from the internet [30]. It uses generative AI to facilitate natural conversations across multiple formats, including text, voice, and images.

The development of chatbots has traditionally focused on languages with extensive NLP resources, such as English, leaving Turkish relatively unexplored. Several studies have explored chatbot development for other languages, but few have addressed the unique challenges of Turkish. This paper aims to fill this gap by introducing TrBot, a deep learning-based general-context chatbot designed specifically for Turkish.

A notable recent development in this field is the adaptation of chatbot technologies to the Turkish language. This innovation addresses the unique linguistic and cultural characteristics of Turkish, which are often underrepresented in mainstream chatbot applications. Incorporating advanced chatbot models with Turkish language capabilities enhances user interactions and engagement for Turkish-speaking users, offering a more tailored and effective conversational experience.

Overall, the literature reviewed highlights the diverse approaches and methodologies in chatbot development, from rule-based systems to deep learning models and reinforcement learning frameworks. Innovations such as TrBot and advancements in Turkish language processing represent significant strides toward creating intelligent and adaptive conversational agents with wide-ranging applications in entertainment, education, customer service, and beyond.

III. MOTIVATION BEYOND THE SEQ2SEQ MODEL FOR THE TURKISH LANGUAGE

With its rich morphological structure and agglutinative nature, the Turkish language presents unique challenges for NLP models. The complex word formations and extensive use of suffixation in Turkish require models that can effectively capture long-term dependencies and handle context-sensitive inflections. This section outlines the motivations for selecting the Seq2Seq model over Transformer architectures for developing a Turkish chatbot.

A. COMPLEX MORPHOLOGY IN TURKISH

Turkish is characterized by its agglutinative nature, where words are formed by adding various suffixes to root words. This leads to a vast array of possible word forms. For example, the root word ''gel'' (come) can generate multiple inflections such as ''geldim'' (I came), ''geleceğim'' (I will come), and ''geliyorum'' (I am coming). These inflections necessitate a model that can comprehend sequential dependencies across extended contexts, a task well-suited for Seq2Seq models.

B. SEQ2SEQ'S STRENGTH IN CAPTURING SEQUENTIAL DEPENDENCIES

The Seq2Seq model, particularly when enhanced with LSTM layers, processes input sequences token by token, retaining memory of previous tokens through hidden states. This sequential processing is particularly effective for:

- Capturing long-term dependencies within sequences.
- Managing word order and suffix-based grammatical changes prevalent in Turkish.

For instance, in the sentence ''*Ben okula gidiyorum*'' (I am going to school), the suffixes modify meanings based on context and sentence structure. The Seq2Seq model learns these dependencies naturally through its step-by-step processing.

C. CHALLENGES WITH TRANSFORMER ARCHITECTURES FOR TURKISH

While Transformer models have revolutionized NLP with their self-attention mechanisms and positional encodings, they face specific challenges when applied to Turkish:

- *Token Granularity*: Sub word tokenization often splits Turkish words into fragments that lose their morphological meaning (e.g., ''*gidiyorum*'' might be tokenized into ''*gid*,'' ''*iyor*,'' and ''*um*''). Seq2Seq models, focusing on sequential word-level dependencies, handle this more effectively.
- *Data Requirements*: Transformers require large datasets to capture nuanced linguistic patterns. The lack of extensive, high-quality Turkish datasets limits the ability of Transformers to generalize well, particularly for complex morphological structures.

D. EMPIRICAL PERFORMANCE

Our experiments demonstrate that the Seq2Seq model outperforms Transformer architectures in accuracy, BLEU score, and perplexity for QA and dialog datasets. This superior performance is attributed to the Seq2Seq model's ability to:

- Maintain grammatical coherence.
- Capture the relationships between morphemes (root words and suffixes) critical for Turkish.

E. MEMORY AND CONTEXT RETENTION

LSTM-based Seq2Seq models inherently retain memory across longer sequences, allowing them to track relationships between words and their morphological changes effectively. In contrast, Transformers, relying on fixed window sizes for attention mechanisms, may truncate context in long Turkish sentences [31].

F. PRACTICAL ADVANTAGES FOR RESOURCE-CONSTRAINED SETTINGS

Seq2Seq models are computationally efficient and well-suited for the dataset sizes currently available for Turkish. In comparison, Transformers require extensive computational resources, which may not be justified given the limitations in available data. This makes Seq2Seq models a practical choice for foundational models in Turkish language chatbot development.

G. FOCUS ON FOUNDATIONAL DEVELOPMENT FOR TURKISH CHATBOTS

While more advanced models like Transformers or BERT [32] are available, the research aimed to establish a foundational model for Turkish-language conversational AI. Given the limited availability of Turkish-specific NLP resources and the significant preprocessing requirements of transformer-based architectures, the Seq2Seq model provides a robust starting point for pioneering efforts in this domain. By developing and fine-tuning TrBot on the proposed datasets, we could lay the groundwork for further advancements.

Thus, the Seq2Seq model's sequential processing, ability to capture long-term dependencies, and better alignment with Turkish morphology make it more suitable for handling the complexities of the language. This approach provides a practical and efficient solution for Turkish language chatbot development, particularly given current constraints in dataset size and computational resources. By focusing on these motivations, it is evident that the Seq2Seq model addresses the linguistic challenges of Turkish more effectively than Transformer architectures, making it the preferred choice for developing robust conversational agents in Turkish under the currently available resources.

IV. METHODOLOGY

The development and evaluation of TrBot, a Turkish deep-learning chatbot, involved several critical phases. These phases encompassed the creation and preparation of datasets, the design and implementation of the model architecture, the training process, and the comprehensive evaluation of the model's performance.

A. DATASETS CREATION AND CHARACTERISTICS

To overcome the lack of suitable Turkish datasets, two comprehensive datasets were developed: question-answer (QA) Dataset and Dialogue Dataset. These datasets were designed to enhance TrBot's factual accuracy and conversational fluency.

1) QA DATASET

Purpose: Train TrBot for accurate and direct question-answer interactions.

Size: 40,702 entries, approximately 67,500 unique words.

Sources: Wikipedia and social media platforms.

Structure: Each entry consists of a question and its corresponding answer. For example:

Question: "buzul mağaralarınasıl oluşur?"

Answer: "Buzul mağarası, bir buzulun buzunun içinde oluşan bir mağaradır."

Domains: Encyclopedic knowledge (e.g., geography, history) and real-world social media queries.

Crafting Process: Questions were generated from Wikipedia headers and key sentences, with answers sourced from summaries or excerpts.

Social media queries were selected using NLP-based topic modeling, and answers were validated using trusted sources.

2) DIALOG DATASET

Purpose: Enhance TrBot's conversational abilities across diverse contexts.

Size: 304,446 entries, approximately 167,000 unique words.

Sources: Movie subtitles and social media conversations.

Structure: Represents a variety of conversational interactions, from casual exchanges to structured dialogues. Example:

Person1: "Bugün hava durumu nasıl?"

Person2: "Dışarısıgüneşli ve sıcak görünüyor!"

Domains: Informal movie dialogues and modern conversational styles from social media.

3) TOOLS FOR DATA COLLECTION

Web Scraping: Tools such as Beautiful Soup, Scrapy, and Selenium were used to extract static and dynamic content.

API Access: Public social media APIs were utilized, adhering to platform terms of service.

Manual Curation: Selected data underwent manual refinement for quality assurance.

4) ETHICAL CONSIDERATIONS

Privacy Protection: The datasets were created using only publicly available data, guaranteeing adherence to all applicable legal and ethical norms. Personal identifiers were anonymized or completely removed at the data processing stage to reduce the risk of disclosing sensitive information.

Compliance: Data collection adhered to platform-specific terms of service.

Bias Mitigation: Dataset biases affecting chatbot fairness include selection bias, where certain demographics or viewpoints are overrepresented, implicit bias where societal stereotypes embedded in data lead to discriminatory outputs, annotation bias arises from subjective labeling by annotators, introducing inconsistencies, while interaction bias occurs when user interactions reinforce existing biases during deployment [33], [34]. These issues collectively result in unfair or offensive behavior. To mitigate biases, this study emphasizes the use of diverse sources to ensure linguistic and cultural inclusivity; however, further investigation is needed to ensure comprehensive bias mitigation, as this is preliminary research exploring foundational approaches.

Risks: The risks of deploying a chatbot in sensitive cultural or social contexts include the reinforcement of harmful stereotypes, preservation of biases such as gender biases, and potentially harmful interactions that can affect societal values and justice [34].

5) DATASET FORMAT AND ANNOTATION

QA Dataset: Stored in CSV/JSON.

Conversation Dataset: Stored in JSON/TXT for hierarchical dialogue representation.

Annotation: Semi-automatic tagging (e.g., domain, sentiment) was performed using NLP tools like spaCy and NLTK, followed by manual validation.

6) VISUALIZATION AND ONLINE AVAILABILITY

Figures 1 and 2 visually depict the structure of the datasets and highlight their comprehensive nature. Figure 1 illustrates the sentence word length frequency in the QA dataset, whereas Figure 2 shows the sentence word length distribution in the conversation dataset. These figures underscore the comprehensive nature of the datasets, which are essential for training a versatile and practical chatbot.

Availability: The datasets used in this study are available on IEEE DataPort with the DOI: 10.21227/1et1-8e90. This ensures transparency, reproducibility, and accessibility for researchers interested in Turkish NLP tasks.

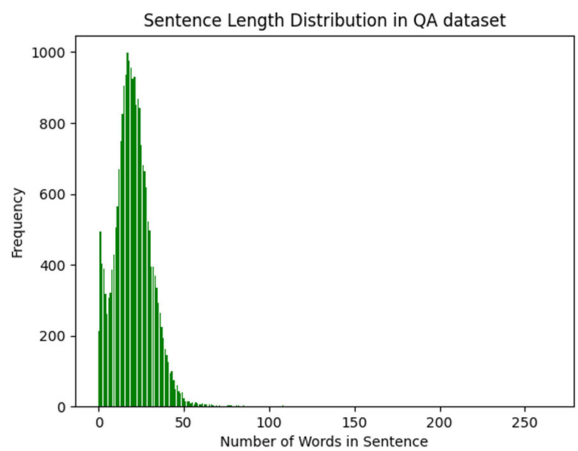

FIGURE 1. Sentence word length frequency in QA dataset.

B. DATA PREPROCESSING

The data preprocessing has five steps, as elaborated below:

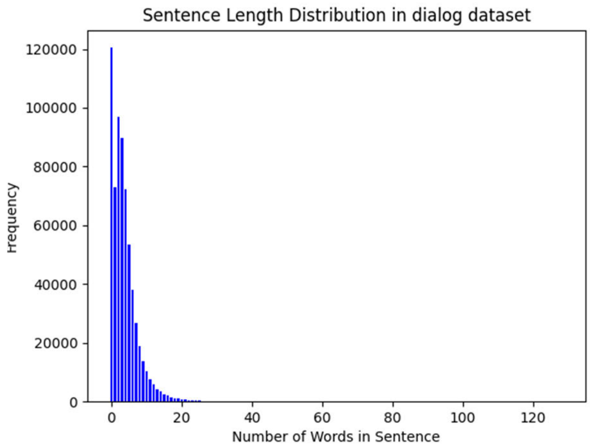

FIGURE 2. **Sentence word length in Dialog dataset.**

1) TOKENIZATION

Tokenization is the process of splitting text into smaller units called tokens, such as words or subwords. For instance, the sentence "Bugün hava güzel" (Today the weather is nice) is split into tokens: ["bugün," "hava," "güzel"]. This step ensures the model can process and understand text data effectively, as it works with these individual tokens during training.

2) LOWERCASING

Text is converted to lowercase to ensure consistency and reduce redundancy in data representation. For example, "Bugün" and "bugün" are treated as the same word after lower casing. This step is especially helpful in languages like Turkish, where capitalization is used extensively, such as in proper nouns or sentence beginnings.

3) REMOVING SPECIAL CHARACTERS AND PUNCTUATION

Special characters, numbers, and punctuation marks that do not contribute to the linguistic meaning are removed to simplify the data. For instance, a sentence like "*Merhaba! Bugün nasılsın?*😊" becomes "*merhaba bugün nasılsın*". This helps reduce noise in the data, allowing the model to focus on meaningful content.

4) HANDLING RARE WORDS AND OUT-OF-VOCABULARY (OOV)

Words that appear infrequently in the dataset (rare words) or are not present in the model's vocabulary (OOV tokens) are significant challenges in NLP. These are addressed through several strategies:

- *Sub word Tokenization*: Breaking down words into smaller components using methods like Byte Pair Encoding (BPE) or Sentence Piece. For example, the word "*güzelmiş*" (it was beautiful) can be split into sub words: ["*güzel*," "*miş*"]. This ensures that even if the full word is unseen, the sub words can still be processed.
- *Replacement with a Special Token*: Rare words or OOV tokens can be replaced with a placeholder like <UNK> (unknown token) to signify their presence without trying to process them directly.
- *Frequency Thresholding*: Words that occur below a specific frequency threshold in the training dataset are excluded from the vocabulary to prevent overfitting on noise. This is particularly useful for Turkish, where agglutinative word formation leads to many rare word forms.
- *Morphological Decomposition*: For Turkish, which is morphologically rich, breaking down rare words into their root forms and suffixes allows the model to understand their meaning. For instance, "*kitapçılardandı*" (it was from the bookstores) can be decomposed into "*kitap*" (book), "*çı*" (suffix indicating profession or place), "*lar*" (plural suffix), and "*dan*" (ablative suffix).

5) LANGUAGE-SPECIFIC PREPROCESSING (TURKISH MORPHOLOGICAL ANALYSIS AND STEMMING)

Morphological analysis involves identifying the root and affixes of words to simplify complex word forms. For example, "*okullarımızdan*" (from our schools) is broken down into:

- Root: "*okul*" (school)
- Suffixes: "*lar*" (plural), "*ımız*" (possessive indicating "our"), and "*dan*" (ablative meaning "from").

Stemming, on the other hand, reduces words to their base forms, which aids in normalizing the text and reducing vocabulary size.

C. MODEL ARCHITECTURE DESIGN

TrBot's model architecture leverages the Seq2Seq framework, enhanced with LSTM units, which are renowned for their capacity to capture long-range dependencies in sequential data [35]. Within this architecture, the encoder processes input sequences, such as user questions, through multiple stacked LSTM layers. These layers encode the input into a fixed-length vector representation, effectively preserving crucial contextual and semantic information from extended sequences.

Following the encoding phase, the decoder network, also composed of stacked LSTM layers, utilizes the encoded representation (context vector) to generate output sequences corresponding to TrBot's responses. The LSTM layers in the decoder ensure the generation of coherent and contextually relevant responses, capturing dependencies between words and phrases within the generated sequence. This approach enables TrBot to navigate the nuances and complexities of the Turkish language, producing accurate and meaningful conversational exchanges.

To handle the rich morphological structure of Turkish, TrBot incorporates three stacked LSTM layers in both the encoder and decoder, each containing 1024 LSTM units.

The architecture also employs dropout regularization during training to minimize overfitting and improve generalization. Shared word embeddings of size 1024 dimensions are used across the encoder and decoder, converting input and output tokens into dense vector representations aligned with the LSTM dimensions.

The sequential processing capabilities of LSTMs are particularly advantageous in handling Turkish's complex grammar and diverse sentence constructions. By maintaining context over long input sequences, TrBot delivers accurate responses, even in scenarios involving intricate sentence structures and varied conversational contexts. This design ensures that users receive natural, context-aware, and linguistically appropriate responses.

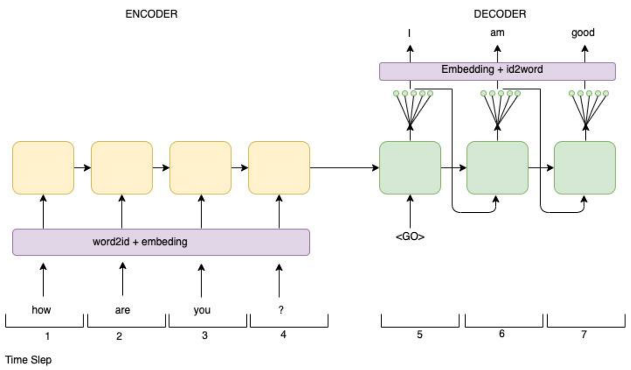

FIGURE 3. Seq2Seq model [36].

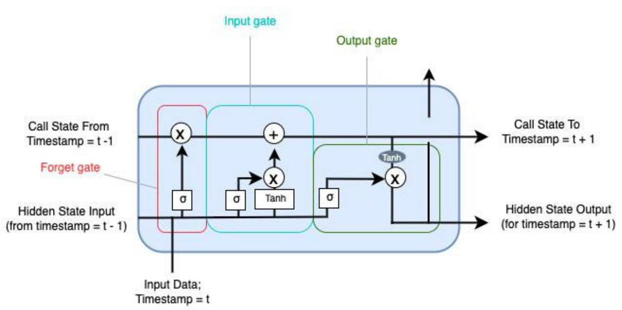

FIGURE 4. Structure of LSTM cell gated [35].

Figures 3 and 4 provide visual representations of the Seq2Seq model and the LSTM architecture, respectively, illustrating the detailed flow of data through the model and highlighting the critical components of the architecture.

The workflow of the TrBot model Architectural Design is shown in Figure 5. It illustrates the systematic transformation from user input to generated responses as:

1) *Input Sequence*: User-provided raw textual data (e.g., questions or conversational inputs).
2) *Encoder*: The input sequences are processed through stacked LSTM units, which transform them into a context vector. This vector encapsulates both syntactic and semantic features of the input.

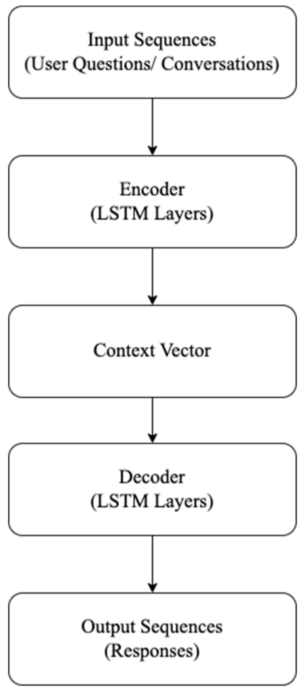

FIGURE 5. TrBot model architectural design.

3) *Context Vector*: The fixed-length representation acts as a bridge, carrying meaningful information from the encoder to the decoder.
4) *Decoder*: The decoder, also composed of LSTM units, processes the context vector to produce output sequences. These responses are crafted word-by-word while maintaining contextual consistency.

D. INCORPORATING TRANSFORMER MODELS FOR TrBOT

To address the evolving advancements in NLP, an additional version of TrBot using Transformer-based architecture was developed. Transformers have revolutionized NLP by enabling parallel processing of input sequences and leveraging attention mechanisms to capture long-range dependencies effectively.

The Transformer-based version of TrBot utilized a standard encoder-decoder architecture, like the model proposed by Vaswani et al [11]. The key components of this architecture are:

- *Multi-Head Self-Attention Mechanism*: This component facilitates better context comprehension by allowing the model to focus on different parts of the input sequence simultaneously.
- *Positional Encoding*: Since transformers do not inherently process sequences in a specific order, positional encoding is employed to help the model retain sequential information, which is crucial for language tasks. This encoding injects information about the relative or absolute position of tokens in the sequence, allowing the model to understand the order of words.

- *Feedforward Neural Networks*: These are integrated within each layer to perform non-linear transformations on the input data. The feedforward networks consist of multiple layers of neurons, enabling the model to learn complex patterns and representations.

The architecture specifics for the Transformer-based TrBot included:

- *Number of Heads*: 8, which allows the model to attend to different parts of the input sequence in parallel, enhancing its ability to capture diverse aspects of the data.
- *Depth*: The model consisted of 6 encoder layers and 6 decoder layers, providing sufficient depth to learn intricate patterns and relationships within the data.

E. TRAINING

The model training process involved feeding both the QA and conversation datasets into the Seq2Seq model and the Transformer model. During training, the model iteratively learned to map input sequences to output sequences, optimizing its parameters to minimize prediction errors. The training was conducted on a virtual machine (VM) equipped with high-performance computing resources, which facilitated efficient model convergence. The training process utilized advanced techniques such as mini-batch gradient descent and backpropagation to update the model parameters. Mini-batch gradient descent allowed the model to learn from subsets of the data, improving the efficiency and stability of the training process. Backpropagation was used to calculate the gradients of the loss function with respect to the model parameters, enabling the model to adjust its weights and biases to better fit the data. Throughout the training, various hyperparameters were carefully tuned, including the learning rate, batch size, and the number of epochs. These hyperparameters were selected to balance the speed of convergence with the accuracy of the model. By leveraging these techniques and resources, the Seq2Seq model was trained to effectively handle the complexities of the Turkish language, resulting in a robust chatbot capable of engaging in meaningful and contextually appropriate conversations.

F. EVALUATION

To evaluate the performance of TrBot, several metrics [37] were employed, including accuracy, BLEU score, and perplexity. Accuracy measures the proportion of correctly predicted responses, while the BLEU score assesses the quality of generated responses by comparing them to reference responses. Perplexity measures the uncertainty of the model in predicting the next word in a sequence. An evaluation was conducted using a held-out validation set, and the results were analyzed to assess TrBot's effectiveness in generating coherent and contextually relevant responses.

1) ACCURACY

Accuracy measures the proportion of correctly predicted responses out of the total number of responses generated by the chatbot and is calculated as follows.

$$Accuracy = \frac{Number\ of\ Correct\ Responses}{Total\ Number\ of\ Responses} \tag{1}$$

This metric provides a straightforward measure of how often TrBot's responses match the expected correct answers. However, it may not capture the quality or relevance of the responses beyond correctness.

2) BLEU (BILINGUAL EVALUATION UNDERSTUDY) SCORE

The BLEU score is a metric used to evaluate the quality of text generated by a model by comparing it to one or more reference texts. It is widely used in machine translation and text generation tasks.

$$BLEU = BP.\exp\left(\sum_{n=1}^{N} w_n log p_n\right) \tag{2}$$

where BP is the brevity penalty, w_n are the weights for different n-gram precisions, and p_n are the n-gram precisions. The BLEU score ranges from 0 to 1, with higher scores indicating better alignment with the reference responses. It considers factors like the precision of n-grams (contiguous sequences of words) in the generated text, making it a robust metric for evaluating response quality.

3) PERPLEXITY

Perplexity is a standard metric used to evaluate the quality of probabilistic language models. It measures how well a model predicts a sequence of tokens, reflecting the model's ability to generate fluent and coherent text. Lower perplexity values signify better performance, indicating that the model assigns higher probabilities to the observed text sequences.

Mathematically, perplexity is defined as follows.

$$Perplexity = 2^{-\frac{1}{N}\sum_{i=1}^{N} log_2(p(x_i))} \tag{3}$$

where $P(x_i)$ is the model's predicted probability for the $i-th$ token in a sequence of N tokens.

4) APPLICATION OF EVALUATION METRICS TO TrBoT

To evaluate TrBot, we conducted experiments using a held-out validation set that was not seen during training. Each response generated by TrBot was compared against the expected responses from the datasets. The evaluation process involved the following steps:

1. *Accuracy Measurement*: Responses were labeled as correct or incorrect based on their alignment with the expected answers. Accuracy was then calculated to determine the proportion of correct responses out of the total number of responses generated.
2. *BLEU Score Calculation*: Generated responses were compared to reference responses using n-gram matching [38]. BLEU scores were calculated to assess the quality of the responses, providing an indication of how closely the generated text matched the reference text.

3. *Perplexity* is particularly useful for comparing the effectiveness of various models across datasets. In this study, we evaluated the perplexity of the chatbot across two datasets: a QA dataset and a Dialogue dataset. The comparison was performed using two models: a Seq2Seq and a Transformer-based architecture.

By employing these evaluation metrics, we were able to quantify TrBot's performance in generating accurate and contextually appropriate responses.

V. EXPERIMENTAL SETUP AND RESULTS

In this study, the experimental setup was conducted on a virtual machine (VM) equipped with 16 virtual CPUs (vCPUs) and 48 GB of memory to ensure computational efficiency. The VM utilized Python, TensorFlow, and Matplotlib for model development and analysis, facilitating seamless implementation and visualization of results.

The datasets used for training and validating TrBot included:

- QA Dataset: Consisting of 40,702 entries with approximately 67,500 unique words.
- Dialog Dataset: Comprising 304,446 entries with around 167,000 unique words.

Each dataset was split into training, validation, and test sets using an 80-10-10 split, the distribution of records is illustrated in Table 1:

- 80% Training: Used to train the Seq2Seq and Transformer models.
- 10% Validation: Held-out subset used for monitoring the model's generalization during training and for hyperparameter tuning.
- 10% Testing: Reserved exclusively for final evaluation to assess the model's performance on unseen data.

TABLE 1. Distribution of records in the QA and dialog datasets.

Dataset	Training	Validation	Test
QA Dataset	32,561	4,070	4,071
Dialog Dataset	243,557	30,445	30,444

The TrBot model was trained with a batch size of 32 across varying epochs, ranging from 100 to 1000, in increments of 100 epochs, allowing flexibility in analyzing convergence and performance trends. Each configuration was run 10 times to ensure robustness.

The learning rate was set at 0.001 to stabilize gradient descent optimization and avoid overshooting. The validation set was used to compute evaluation metrics like accuracy, BLEU score, and perplexity during training. Final testing was performed on the unseen test set to provide an unbiased evaluation of the model's effectiveness.

Figure 6 illustrates the training time comparison between Seq2Seq and Transformer-based models on QA and Dialog datasets over 1000 epochs. While Seq2Seq models exhibit higher training times—reaching approximately 30,000 seconds for the Dialog dataset and 10,000 seconds for the QA dataset—they remain a robust choice for NLP tasks, particularly for the morphological structure of the Turkish language. Despite the higher computational efficiency of Transformer-based models, the Seq2Seq architecture demonstrates better results for Turkish, leveraging its ability to effectively model the language's complex morphology.

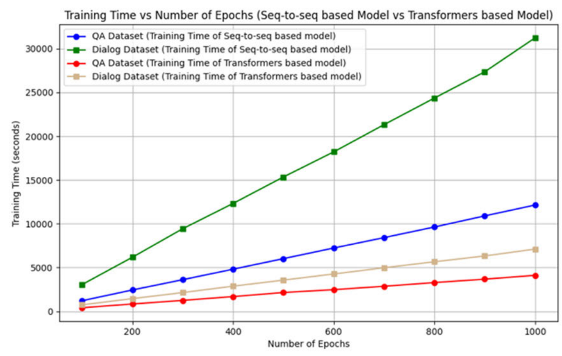

FIGURE 6. Total time taken in each run on both datasets for Seq2Seq and Transformer models.

This analysis underscores the importance of Seq2Seq models beyond training efficiency, as they excel in generating high-quality outputs for Turkish-specific tasks. While training time is a factor, their performance on key measures such as accuracy, BLEU score, and perplexity (discussed in subsequent sections) further highlights their suitability for this context.

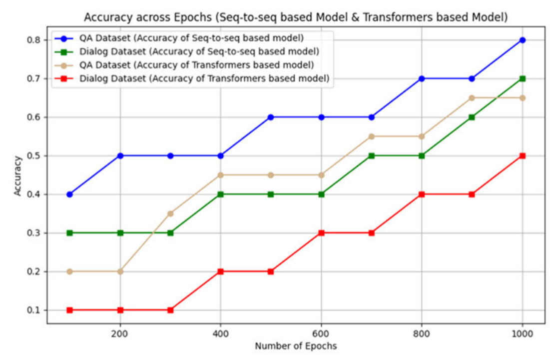

FIGURE 7. Accuracy vs epochs on both datasets for Seq2Seq and Transformer models.

Figure 7 compares the accuracy of Seq2Seq and Transformer-based models across different numbers of epochs for two distinct datasets: the QA dataset and the Dialog dataset. For the QA dataset, Seq2Seq models achieve significantly higher accuracy, reaching approximately 0.8 by the 1000th epoch. On the Dialog dataset, Seq2Seq models also show superior performance, achieving an accuracy of around 0.7 at the same point. In contrast, Transformer-based models exhibit more modest accuracy improvements, with a peak accuracy of 0.6 for the QA dataset and 0.5 for the Dialog dataset.

These findings emphasize the effectiveness of Seq2Seq models, particularly for tasks involving structured question-answering and conversation generation. The Seq2Seq models excel in handling the sequential dependencies and morphological intricacies, such as those found in the Turkish language, which are crucial for both QA and conversational tasks. While Transformer models are known for their computational efficiency, the higher accuracy achieved by Seq2Seq models underlines their strong suitability for achieving reliable language understanding and generation, especially in more complex or context-dependent tasks like dialogue generation and answering questions.

Figures 8, 9, 10, and 11 illustrate the performance of two different models—Seq2Seq model and Transformers-based model—on Dialog and QA datasets over increasing training epochs, tracking correct and incorrect responses. Both models show improvement in performance with training, characterized by an increase in correct responses (green line) and a decrease in incorrect responses (red line). However, notable differences in learning dynamics and overall accuracy emerge when comparing their behavior.

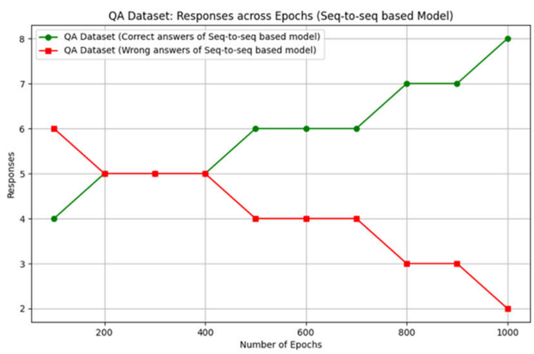

FIGURE 8. **Correct vs Wrong answers over Epochs on QA-Dataset for seq2seq based model.**

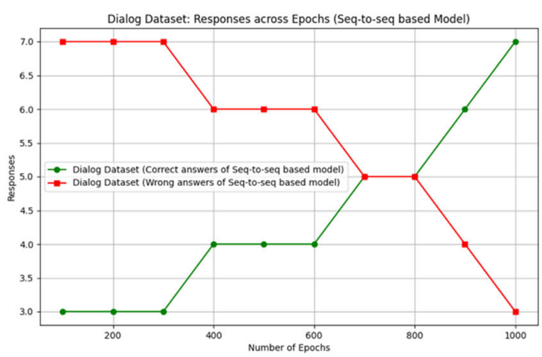

FIGURE 9. **Correct vs wrong answers over epochs on dialog-dataset for seq2seq based model.**

For the Dialog dataset, the Seq2Seq model demonstrates faster convergence, with correct and incorrect responses meeting around 600 epochs and reaching a peak of 7 correct and 3 incorrect responses by epoch 1000. In contrast, the Transformers-based model shows slower progress, with correct responses increasing to 5 and incorrect responses decreasing to 5 by epoch 1000. This indicates a less steep learning curve for the Transformers-based model, suggesting it requires more epochs or additional optimization to achieve similar accuracy levels.

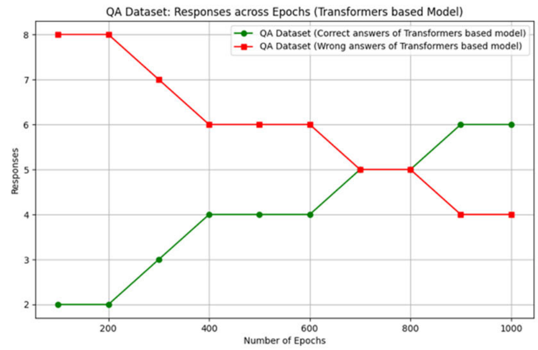

FIGURE 10. **Correct vs Wrong answers over Epochs on QA-Dataset for Transformers-based model.**

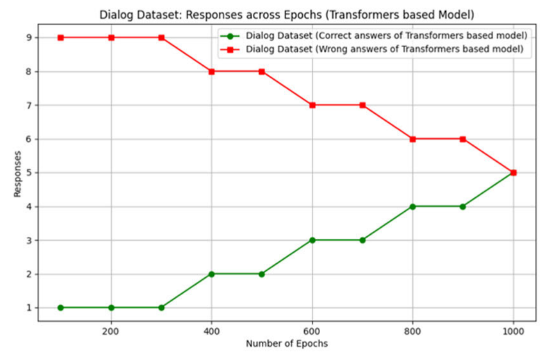

FIGURE 11. **Correct vs Wrong answers over Epochs on Dialog-Dataset for Transformers based model.**

For the QA dataset, the Seq2Seq model outperforms the Transformers-based model both in learning speed and final accuracy. The Seq2Seq model achieves near-perfect accuracy by epoch 1000, with 8 correct and 2 incorrect responses. Conversely, the Transformers-based model shows moderate improvement, reaching 6 correct and 4 incorrect responses by the same point. This comparison highlights the Seq2Seq model's superior adaptability and efficiency in learning patterns from the QA dataset.

An additional advantage of the Seq2Seq model lies in its capacity to handle languages with complex morphology, such as Turkish. Turkish presents unique challenges due to its agglutinative structure and extensive morphological richness, which require models to capture long dependencies and contextual nuances effectively. The Seq2Seq model's architecture is particularly well-suited for these tasks, as it excels at modeling intricate linguistic structures and generating contextually appropriate responses. Its faster convergence and higher accuracy in the provided datasets further reinforce its potential for Turkish natural language processing tasks, making it a compelling choice for researchers and developers working in this domain.

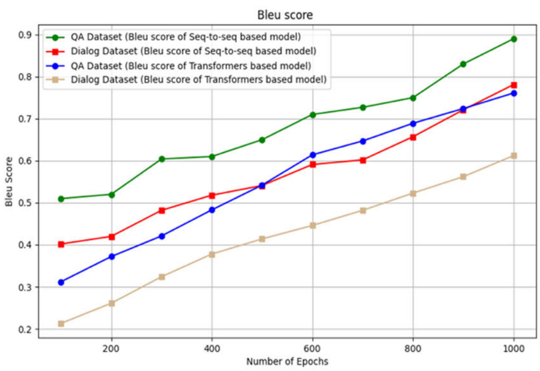

FIGURE 12. **BLEU Score on both datasets for Seq2Seq and Transformer models.**

The BLEU score results, as shown in Figure 12, provide additional insights into model performance over epochs. The Seq2Seq model achieves consistently higher BLEU scores compared to the Transformers-based model across both datasets. For the QA dataset, the Seq2Seq model's BLEU score increases from 0.5 at 200 epochs to approximately 0.9 at 1000 epochs, reflecting its superior ability to generate accurate and contextually relevant responses. On the Dialog dataset, the Seq2Seq model also shows significant improvement, with BLEU scores rising from 0.4 to 0.7 over the same training period. In contrast, the Transformers-based model demonstrates slower progression, achieving BLEU scores of 0.6 and 0.5 for the QA and Dialog datasets, respectively, by epoch 1000. These results reinforce the Seq2Seq model's adaptability and effectiveness, particularly for tasks requiring high linguistic precision.

Figure 13 illustrates the performance of Seq2Seq and Transformer models in terms of perplexity across epochs (100 to 1000) for two datasets: QA and Dialogue datasets. The Seq2Seq model consistently achieves lower perplexity compared to Transformers for both datasets, indicating superior language modeling and predictive performance. Notably, the perplexity for the QA dataset is lower than for the Dialogue dataset across all models, reflecting the structured and predictable nature of QA data compared to the more diverse and complex conversational data.

Perplexity decreases with increasing epochs for all configurations, showing the models' learning progression. The Seq2Seq model converges faster (around 600 epochs) and maintain better performance than Transformers, particularly excelling in the QA dataset, with a final perplexity of 50 compared to the Transformer's 95 at 1000 epochs. These results highlight Seq2Seq models as the more effective choice for chatbot tasks in this study, especially for structured question-answering scenarios, while Transformers

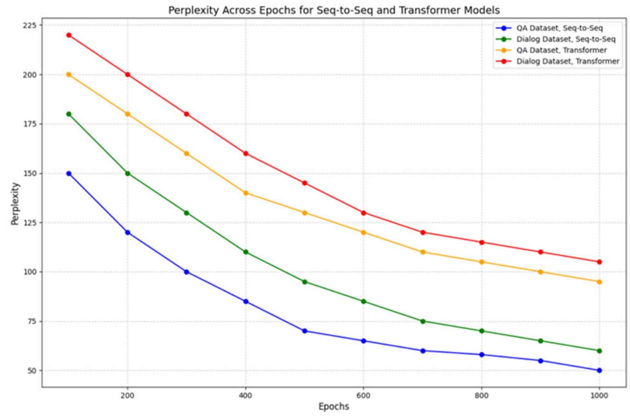

FIGURE 13. **Perplexity Score on both datasets for Seq2Seq and Transformer models.**

may require additional tuning to handle conversational tasks effectively.

Table 2 summarizes a comparison of performance metrics for the two architectures: Seq2Seq model and Transformer model. For the QA dataset, the Seq2Seq model achieved an accuracy of 80%, significantly outperforming the Transformer-based model, which achieved an accuracy of 60%. Similarly, in the dialog dataset, the Seq2Seq model reached an accuracy of 70%, compared to the 50% accuracy of the Transformer-based model. Furthermore, the BLEU scores indicate superior performance by the Seq2Seq model, with scores of 0.90 and 0.77 for the QA and dialog datasets, respectively, compared to 0.76 and 0.61 for the Transformer model. However, it is noteworthy that the Transformer model demonstrated a considerable reduction in epoch time, approximately 70% less than the Seq2Seq model, which has a longer training duration due to the sequential processing nature of LSTM layers.

TABLE 2. **Comparison between transformer-based and Seq2Seq TrBot.**

Metric	Transformer	Seq2Seq
Accuracy (QA Dataset)	60%	80%
Accuracy (Dialog Dataset)	50%	70%
BLEU Score (QA Dataset)	0.76	0.90
BLEU Score (Dialog Dataset)	0.61	0.77
Perplexity (QA Dataset)	95	50
Perplexity (Dialog Dataset)	105	60
Epoch Time	~70% reduction compared to Seq2Seq	Longer due to sequential LSTM processing

In addition to the quantitative analysis, we conducted a qualitative comparison with other state-of-the-art models, including Transformer-based architectures, BERT, and ChatGPT. Implementing and training each of these models for a direct comparison with our Seq2Seq model was deemed impractical due to resource constraints and the extensive computational requirements. Therefore, we employed a

TABLE 3. Comparing the Seq2Seq model used in TrBot with other advanced models like Transformers, BERT, and ChatGPT.

Feature	Seq2Seq (TrBoT)	Transformers	BERT	ChatGPT
Architecture	LSTM-based encoder-decoder model.	Self-attention-based mechanism.	Bidirectional Transformer encoder.	Generative Pretrained Transformer architecture.
Context Handling	Sequential processing, limited long-term context understanding.	Handles long-range dependencies efficiently.	Focuses on bidirectional context for understanding sentences.	Optimized for generating coherent, multi-turn conversations.
Language Support	Turkish-specific.	Multilingual, widely used for NLP tasks.	Pretrained primarily on multilingual corpora.	Pretrained on diverse, multilingual datasets.
Dataset	QA and dialog datasets (proposed in this study).	Large, diverse datasets (e.g., Wikipedia, books).	Wikipedia and BookCorpus.	Extensive datasets including conversational data.
Performance	80% QA accuracy, 70% dialog accuracy; BLEU scores up to 0.9.	State-of-the-art across NLP tasks with fine-tuning.	High performance in language understanding and classification.	Excels in conversational and generative tasks.
Training Efficiency	High computational requirements for LSTM-based models.	More parallelizable than LSTMs due to attention mechanisms.	Efficient for tasks requiring understanding, but pretraining is expensive.	Computationally intensive but optimized for text generation.
Scalability	Limited by dataset size.	Scalable to large datasets with pretraining.	Scalable but primarily for understanding, not generation.	Highly scalable with transfer learning for generative tasks.
Applications	Basic QA and dialog systems.	Versatile across translation, summarization, and more.	QA, sentiment analysis, text classification.	Advanced chatbots, text generation, and content summarization
Adaptability	Limited to Turkish datasets.	Flexible for fine-tuning to various tasks.	Specialized for understanding, requires additional setup for generation.	Highly adaptable for diverse conversational contexts.
Innovations	Application of Seq2Seq to Turkish with custom datasets.	Revolutionized NLP with attention mechanisms.	Introduced bidirectional attention for better comprehension	Breakthroughs in conversational AI with fine-tuned dialogue.

descriptive comparative approach to evaluate their theoretical and practical capabilities. Our qualitative analysis, shown in Table 3, provides valuable insights into the relative strengths and weaknesses of each model.

Error Analysis: During the evaluation, we observed that TrBot occasionally struggles with complex morphological structures in Turkish, such as handling long agglutinative words or context-dependent suffixes. For instance, in response to the input "Bugün hava nasıl?" (How is the weather today?), TrBot generated the output "Hava güneşli." (The weather is sunny.), which, while correct, lacked additional details such as temperature or context. These limitations highlight the challenges posed by Turkish morphology and the need for further improvements in dataset diversity and model architecture.

VI. DISCUSSION

The development and evaluation of TrBot represent a pioneering step in the field of NLP, particularly for Turkish, a language with complex morphological structures. The results indicate that TrBot performs well, achieving 80% accuracy on the QA dataset and 70% on the dialog dataset, underscoring its potential as a conversational agent.

TrBot's development marks a groundbreaking step in the creation of Turkish-language conversational agents. Prior to TrBot, there were limited efforts to develop chatbots tailored specifically to the Turkish language. The absence of Turkish-specific chatbots may be attributed to the complexity of Turkish grammar and morphology, which poses challenges for NLP models typically trained on English or other widely spoken languages.

While the Transformer model exhibited reduced training times due to its parallelizable architecture, its performance in terms of accuracy and BLEU scores was consistently lower than that of the Seq2Seq model. This outcome underscores several key points:

1) *Data Size vs. Model Complexity*: Transformers typically require significantly larger datasets to achieve their full potential. Although our QA and dialog datasets are substantial within the context of Turkish-language research, they were likely insufficient for the Transformer model to excel, particularly in handling the complex morphology of Turkish.
2) *Contextual Understanding*: The Seq2Seq model with LSTM layers demonstrated superior contextual understanding and coherence in generating responses. This can be attributed to its ability to effectively capture sequential dependencies, which is crucial for morphologically rich languages like Turkish.
3) *Resource Utilization*: While Transformers are computationally efficient during training due to their parallel processing capabilities, their higher memory demands and greater complexity make them less practical for deployment in resource-constrained environments compared to Seq2Seq models.
4) *Trade-offs Between Models*: The lower performance of the Transformer model highlights the necessity for larger datasets and potentially more advanced fine-tuning strategies to handle the nuances of Turkish conversational AI effectively.

A. RESEARCH CHALLENGES

Despite its achievements, the development of TrBot encountered several significant challenges. One of the primary obstacles was the creation of suitable datasets. Unlike English, for which large-scale datasets are readily available. Additionally, while TrBot's performance is a promising foundation for further development, there remains considerable room for improvement, particularly in the dialog dataset. This suggests that although TrBot is effective in handling structured QA formats, it struggles with the more nuanced and context-dependent nature of conversational interactions. Such limitations are common in chatbot development, especially for languages with complex grammatical structures like Turkish. These issues may stem from the difficulties of

Turkish morphology, the limited size of the training datasets, and the challenges inherent in capturing contextual details within conversations, and computational limitations that restricted the scope of experiments, preventing exhaustive hyperparameter tuning and extended training periods.

To address the model's limitations, particularly the reduced accuracy on the dialog dataset, several strategies can be implemented. Expanding the training datasets by incorporating more diverse and extensive Turkish language resources is a critical first step. This expansion would enhance the model's ability to generalize and perform better in varied conversational contexts. While our study found that the Seq2Seq model outperforms the Transformer-based model in the current setting, it is essential to recognize that the full potential of transformer-based architectures, such as BERT or GPT, may not have been realized due to several factors, primarily arising from hyperparameter tuning and dataset suitability. Transformers require extensive hyperparameter optimization which was constrained by the available computational resources in this study. Additionally, the relatively small size and limited diversity of the Turkish-specific datasets reduced the model's ability to generalize and leverage its attention mechanism effectively for capturing complex morphological dependencies. These more recent architectures have demonstrated significant promise in other studies, particularly in capturing context and understanding language nuances.

The TrBot's ability to handle Turkish's agglutinative nature demonstrates its potential adaptability to other morphologically rich languages such as Finnish, Hungarian, or Japanese. These languages share challenges such as complex suffixation and long dependencies. Moreover, the study highlights strategies like morphological decomposition and dataset curation that can be generalized to NLP development in low-resource languages, paving the way for better language support globally. Furthermore, TrBot's real-world applications span customer service, education, and healthcare, offering scalable and efficient solutions in Turkish. These applications highlight its societal impact by improving accessibility, promoting digital inclusion, and bridging language barriers for Turkish speakers. Additionally, TrBot supports the digitization of Turkish, fostering cultural representation, while also creating economic opportunities for businesses to better serve Turkish-speaking populations, bridging communication gaps and supporting linguistic diversity in technology.

B. FUTURE DIRECTIONS

To further enhance TrBot's performance and capabilities, several promising research avenues can be explored. These directions focus on expanding the training data, leveraging advanced model architectures, and incorporating new techniques to improve TrBot's ability to engage in natural and contextually appropriate conversations. However, it is important to acknowledge that computational constraints have limited the scope of experiments conducted in this study. An investment in computational resources is needed to handle the following key areas, facilitating extended training periods and broader experimentation. Below are key areas that offer potential for significant advancements:

1) *Dataset Expansion*: Continuously expanding the datasets to improve the variety and depth of conversational contexts. Incorporating data from diverse sources, such as user interactions and domain-specific conversations, will enhance the model's robustness and versatility.
2) *Advanced Architectures*: Experimenting with advanced architectures, such as transformers and attention mechanisms, could significantly improve TrBot's ability to handle the complexities of Turkish conversation. Additionally, exploring hybrid approaches that combine the strengths of Seq2Seq models and transformer-based architectures may provide a more balanced solution.
3) *Reinforcement Learning*: Exploring reinforcement learning techniques could allow TrBot to learn dynamically from interactions, thereby improving its responsiveness and accuracy over time.
4) *Multimodal Inputs*: Integrating multimodal inputs (e.g., text, voice) and considering contextual factors such as user sentiment and intent could further refine TrBot's conversational abilities, making it more adaptable and user-friendly.
5) *User Evaluation*: In the next phase of this study, we aim to incorporate usability evaluation metrics to further assess and improve the TrBot chatbot system. Specifically, the System Usability Scale (SUS) will be utilized to measure the chatbot's usability from an end-user perspective. The SUS, a reliable and widely adopted tool for usability testing, will enable us to capture users' subjective feedback on TrBot's performance, interaction quality, and ease of use. Conducting SUS-based evaluations with participants from diverse demographics will provide quantitative insights into user satisfaction, facilitating targeted improvements to the chatbot's interface, interaction logic, and overall user experience.
6) *Integration of Pre-trained Models:* While BERTurk and XLM-R are highly effective for various NLP tasks, they are primarily designed for language understanding rather than conversational generation. In future work, we plan to explore hybrid models that combine the strengths of Seq2Seq architectures with pre-trained language models like BERTurk to enhance TrBot's performance in understanding and generating Turkish text.
7) *Scalability and Deployment:* In future work, we plan to explore lightweight model architectures, such as model distillation or quantization, to optimize TrBot for resource-constrained environments. Additionally, we will investigate deployment strategies, such as cloud-based solutions or edge computing, to ensure scalability and efficiency in real-world applications.
8) *Model and Hyperparameter Optimization:* While the current study did not include hyperparameter tuning

for the Transformer model, we plan to explore advanced optimization techniques, such as fine-tuning, advanced tokenization, and systematic experimentation with different hyperparameter configurations, to fully leverage the capabilities of Transformer-based architectures. Additionally, optimizing the model for efficiency through techniques such as model pruning and quantization will be crucial to improving TrBot's performance and scalability, particularly for resource-constrained environments.

VII. CONCLUSION

TrBot represents a significant advancement in developing a deep learning-based chatbot capable of engaging users in natural conversations across a wide range of topics in Turkish. This study has detailed the comprehensive development process, including the creation and curation of two extensive datasets, the design of the model architecture, and the rigorous training and evaluation procedures. TrBot's performance, achieving 80% accuracy on the QA dataset and 70% on the dialog dataset, demonstrates its potential as a robust conversational agent for the Turkish language. The results underscore TrBot's significance in the field of NLP for Turkish, particularly given the language's complex morphological structures and the scarcity of large-scale Turkish-language datasets. Our comparative analysis between the Seq2Seq and Transformer-based models revealed that the Seq2Seq model currently outperforms the Transformer model, highlighting the importance of selecting appropriate architectures for specific linguistic challenges. While TrBot shows promise, there remains room for improvement, particularly in handling nuanced and context-dependent conversational interactions. This limitation reflects broader challenges in chatbot development for languages with intricate grammar like Turkish. Overall, while TrBot has demonstrated significant potential, ongoing efforts to enhance its capabilities through better data resources and refined model architecture will be essential. TrBot stands as a noteworthy contribution to Turkish NLP, laying a strong foundation for future advancements in conversational AI.

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